IFEANYI CHRISTOPHER NWAGBARA

Backend Engineer | Python · Django · Flask · FastAPI· PostgreSQL · APIs & Integrations

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SUMMARY

Backend Engineer with 7+ years of experience building and maintaining production Python services, including REST APIs, PostgreSQL-backed systems, background jobs, and third-party integrations. Currently leading backend delivery for an energy-monitoring SaaS, working across Django APIs, telemetry pipelines, multi-tenant access control, and asynchronous processing. Also designed and deployed an open-source digital health follow-up service for hypertension and diabetes programs, covering patient records, blood pressure tracking, appointments, call outcomes, and SMS workflows. Experienced in database performance optimization, CI/CD, code review, and mentoring, with a strong focus on building reliable, maintainable backend systems for real-world operational needs.

PROFESSIONAL EXPERIENCE

Lead Backend Engineer — Wyre Nigeria Limited, Lagos

September 2023 – Present

- Own backend delivery for a multi-tenant energy-management SaaS: REST API design, tenant isolation, role-based access control, schema and migrations, and the production behaviour of the service.
- Build PostgreSQL-backed analytics over large telemetry workloads; cut response times through indexing, execution-plan analysis, database-side aggregation, precomputed and cached read paths, batching, and removal of N+1 queries.
- Run asynchronous workflows on Celery and Redis — scheduled jobs, retries with backoff, idempotency keys, and concurrency controls — so repeated or overlapping work cannot corrupt state.
- Integrate third-party providers and keep them from taking the platform down: timeouts, retries, and degraded-mode fallbacks when an upstream is unavailable.
- Support production: monitoring dashboards and alerting, log-driven investigation, incident response, and plain-language write-ups for product and operations stakeholders.
- Review code, mentor engineers, and document architecture decisions. Also shipped the product's assistant and semantic-search features (OpenAI, pgvector).

Team Lead (Software Development) — Mitiget Assurance and Technology Services Limited, Ikeja, Lagos

June 2020 – August 2023

- Built and maintained Django / Django REST Framework services behind a React front end for governance-and-compliance and e-commerce products.
- Architected a payments and reconciliation platform integrating PayPal and Flutterwave: webhook ingestion, idempotent transaction handling, retry logic, and settlement reconciliation against provider reports.
- Introduced CI/CD pipelines, reusable service templates, and structured code review across a team of five developers, substantially cutting review turnaround.
- Led and mentored five engineers through Agile delivery, task allocation, feedback, and shared engineering standards.
- Added compliance-insight features on OpenAI APIs for executive reporting workflows.

Software Engineer — Sabrud Consortium Nig Ltd, Awka

September 2017 – May 2020

- Developed and maintained backend services for a SaaS energy-management platform using Python, Flask, and MySQL.
- Built a wallet and transaction-tracking system with an auditable ledger, roughly halving billing-related complaints.
- Automated operational workflows and added monitoring, contributing to about a 30% reduction in service downtime.
- Hardened data access with input validation and ORM practice, closing SQL-injection and data-integrity gaps.

NOTABLE PROJECTS

Followup-AI — open-source AI follow-up layer for hypertension and diabetes programs
Stack: FastAPI · PostgreSQL · SQLAlchemy · Alembic · Celery · Redis · React · scikit-learn
***Links**: clinic.1960heritage.com · github.com/IfeanyiXtopher/D_Clinic · MIT*

- Designed a Simple-compatible data model and API surface (patients, blood pressures, appointment chatbot, call results) so the service can sit beside an existing hypertension program without a separate schema translation layer.
- Shipped the core clinical-ops loop: risk-ranked facility worklists under a call budget, an appointment SMS dialogue (confirm / reschedule / stop), identity-free case briefings with factuality checks, and protocol Q&A with citations.
- Deployed as a production-style split system (API + Postgres on a VPS; UI on Vercel), with CI, evaluation reports, and a hard privacy boundary: patient identity never enters a model prompt.

DocCite — Private document Q&A with citations
Stack: Python, FastAPI, LangChain, ChromaDB, OpenAI (text-embedding-3-small, gpt-4o-mini), PyMuPDF, python-docx, React (Vite), SQLite
***Links**: github.com/ifeanyixtopher/DocCite_AI · doccite.1960heritage.com*

- Built an end-to-end RAG system for uploaded documents (PDF, DOCX, TXT, MD): extract text with page metadata, chunk with overlap, embed in batches, and persist vectors in Chroma so the library survives restarts.
- Enforced a strict grounding policy: retrieve top-k chunks above a similarity threshold, prompt only with those excerpts, and refuse when evidence is missing — so answers stay tied to the user's files instead of model guesswork.
- Attached citations from retriever metadata (filename, page when available, short quote), not from free-form model output.
- Shipped it as a deployable product: FastAPI multi-user API (guest TTL + accounts), React UI on Vercel, API on a VPS behind nginx/TLS, per-document summaries, chat-scoped libraries, OpenAI speech-to-text input, and tests around retrieval and refusal behavior.

DaySignal — timezone-aware weather alert delivery
Stack: Ruby on Rails · PostgreSQL · Redis · Sidekiq · Action Mailer · OpenWeatherMap · Mapbox
***Links**: github.com/ifeanyixtopher/DaySignal · daysignal.1960heritage.com*

- Web app for a saved place, a local alert time, and a rain threshold. When that hour arrives and rain risk is high enough, the user gets one email.
- An hourly job selects only the users whose timezone is in that hour, fetches the forecast, and applies the threshold. A same-day record stops a second email for that local date.
- Sign-up, alert preferences, and a map of current conditions are part of the same Rails app. Forecast fetch, the alert decision, and email delivery are separate steps.

TECHNICAL SKILLS

- **Languages:** Python, SQL, JavaScript, Ruby (project experience)
- **Backend:** FastAPI, Django, Django REST Framework, Flask, SQLAlchemy, Alembic, REST API design, pytest, Ruby on Rails (independent project), ActiveRecord
- **Data:** PostgreSQL (schema design, indexing, EXPLAIN / execution-plan analysis, transactions), MySQL, Redis (cache and broker), ETL and data pipelines
- **Async:** Celery, Sidekiq, Sidekiq-Cron, scheduled jobs, retries and backoff, idempotency, concurrency control
- **Operations:** Docker and Docker Compose, GitHub Actions CI/CD, Prometheus and Grafana, structured logging, production incident response
- **Integrations:** Webhooks, payment providers (PayPal, Flutterwave), reconciliation pipelines, third-party API clients (including weather APIs), Action Mailer
- **ML (supporting):** scikit-learn, MLflow, model cards and fairness audits, OpenAI APIs
- **Practices:** Code review, mentoring, Agile delivery, requirements clarification, open-source development

EDUCATION

Bachelor of Engineering (B.Eng.), Electronic and Computer Engineering
University of Nigeria, Nsukka — 2014